

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
Maximum Marks	4 Marks

Technical Architecture:

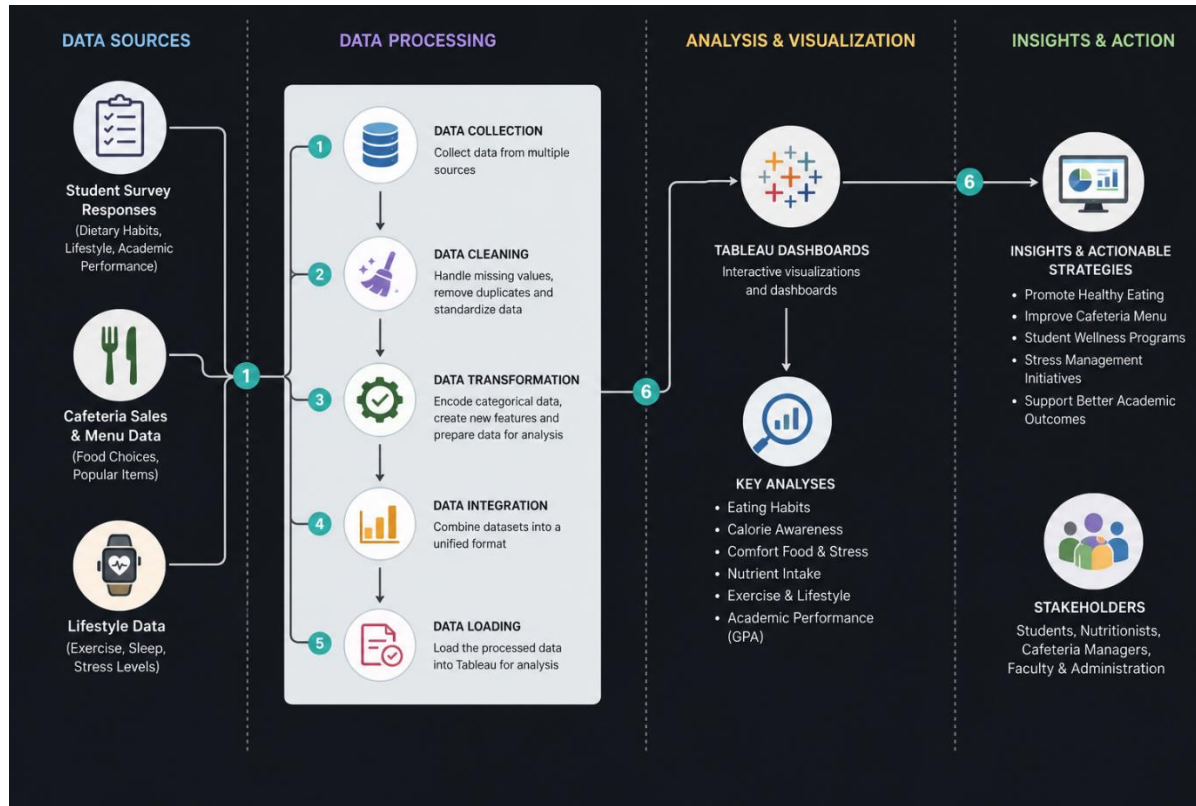


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interactive Tableau dashboards for exploring dietary insights	Tableau Dashboard

2	Application Logic-1	Data cleaning and preprocessing	Python (Pandas, NumPy)
3	Application Logic-2	Data transformation and feature engineering	Python
4	Application Logic-3	Interactive visual analytics and KPI calculations	Tableau
5	Database	College food survey dataset	CSV Dataset
6	Data Repository	Processed analytical dataset	Local Storage / CSV
7	File Storage	Store raw and cleaned datasets	Local File System
8	External Data Source	Student food survey dataset	Kaggle / Public Dataset
9	Visualization Engine	Charts, dashboards and filters	Tableau
10	Analytics Model	Descriptive analytics and trend analysis	Tableau Calculations
11	Infrastructure	Local development and Tableau Desktop	Windows PC, Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Libraries used for preprocessing	Python, Pandas, NumPy

2	Security	Local access and dataset integrity	OS permissions, Tableau project security
3	Scalable Architecture	Modular ETL and dashboard workflow	Python + Tableau
4	Availability	Offline desktop analytics	Tableau Desktop
5	Performance	Optimized extracts, filters and aggregations	Tableau Extracts, Calculated Fields